

6 (a) (i)	arrow in upward direction, foot near P	B1	
(ii)	curved path consistent with (i) between plates	B1	
	then straight (with no kink at change-over)	B1	[3]
(b)	$E = V/d$	C1	
	$= 400 / (0.8 \times 10^{-2})$		
	$= 5.0 \times 10^4 \text{ V m}^{-1}$ (allow 1 sig fig)	A1	[2]
(c) (i)	$F = Eq$	C1	
	$= 5.0 \times 10^4 \times 1.6 \times 10^{-19}$		
	$= 8.0 \times 10^{-15} \text{ N}$ (allow 1 sig fig and e.c.f.)	A1	
(ii)	$a = F/m$	C1	
	$= (8.0 \times 10^{-15}) / (9.1 \times 10^{-31})$		
	$= 8.8 \times 10^{15} \text{ m s}^{-2}$ (allow 1 sig fig and e.c.f.)	A1	[4]
(d)	because F_E is normal to <u>horizontal</u> motion	M1	
	no effect	A1	[2]